

Chapter 15

Body Fluids and Circulation

Key Concept

Transport is essential for maintaining life. Circulatory system ensures distribution of materials throughout the body.

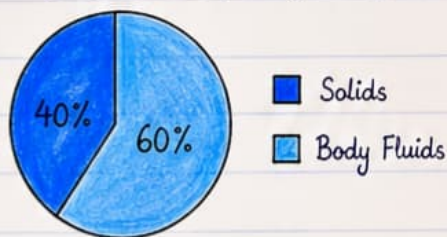
1. Body Fluids

Water is the major component of body fluids. These fluids perform various functions like transport, regulation and protection.

★ Total body weight = ~ 60 kg (in an average adult)

Body fluids = ~ 60% of total body weight.

Distribution of Body Fluids



Compartments of Body Fluids

- ★ Intracellular fluid (ICF) → Present inside the cells ($\frac{2}{3}$ rd of total body fluid)
- ★ Extracellular fluid (ECF) → Present outside the cells ($\frac{1}{3}$ rd of total body fluid)

1. Intravascular fluid (Plasma)

Fluid part of blood, present within blood vessels.

~ 5-6 litres (8% of body weight)

2. Interstitial fluid

Present in the spaces between cells and tissues.

~ 11-12 litres (20% of body weight)

Composition of Plasma

- Water → 90-92%
- Proteins → 6-8%
- Other solutes → 1-2%

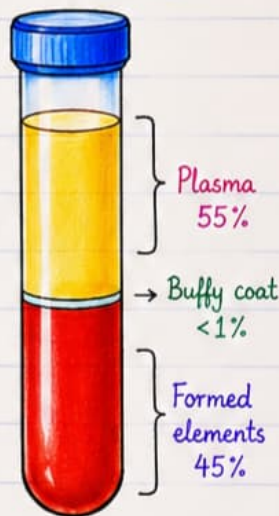
(Glucose, amino acids, hormones, salts, urea, etc.)

3. Transcellular fluid

Present in special areas like CSF, synovial fluid, pleural fluid, peritoneal fluid, etc.

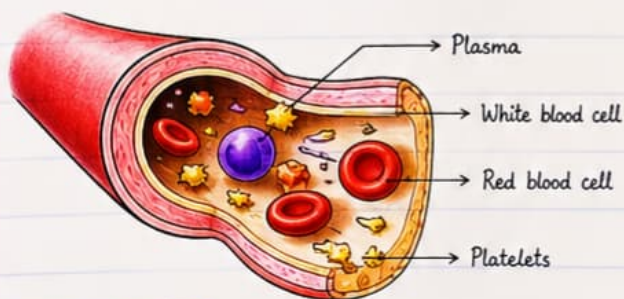
~ 1-2 litres (1-2% of body weight)

Composition of Blood



2. Composition of Blood

Blood is a connective tissue. It is composed of plasma and formed elements.



Plasma → Straw coloured liquid part (55%).

Formed elements (45%) →

- RBCs (Erythrocytes) → Biconcave, enucleate, contain haemoglobin, function in O_2 transport.
- WBCs (Leucocytes) → Nucleated, colorless, play role in defence & immunity.
- Platelets (Thrombocytes) → Cell fragments, help in blood clotting.

Fun Fact

A healthy heart pumps ~ 7,000 litres of blood every day!

Remember

- ICF : ECF = 2 : 1
- Plasma : Formed elements = 55 : 45
- Blood is the only fluid connective tissue in our body.

2. Composition of Blood (Contd.)

Functions of Blood

- **Transport** – of gases, nutrients, hormones, wastes, etc.
- **Regulation** – of pH, body temperature, water balance.
- **Protection** – WBCs provide immunity, platelets help in clotting.

Formed Elements of Blood

1. Red Blood Cells (RBCs).



- Biconcave, non-nucleated
- Contain haemoglobin which transports O_2 .
- Life span – ~120 days

2. White Blood Cells (WBCs)



- Nucleated, colourless
- Defend the body against pathogens.
- Types – Neutrophils, Eosinophils, Basophils, Lymphocytes, Monocytes

3. Platelets (Thrombocytes)



- Small, irregular cell fragments
- Help in blood clotting
- Life span – 5 – 10 days

Blood Clotting (Briefly)

Injury to blood vessel
 ↓
 Platelets release **thromboplastin**
 ↓
 With Ca^{2+} and other factors,
 prothrombin → **thrombin**
 ↓
 Fibrinogen → **fibrin** (insoluble threads)
 ↓
 Forms a clot and prevents blood loss.

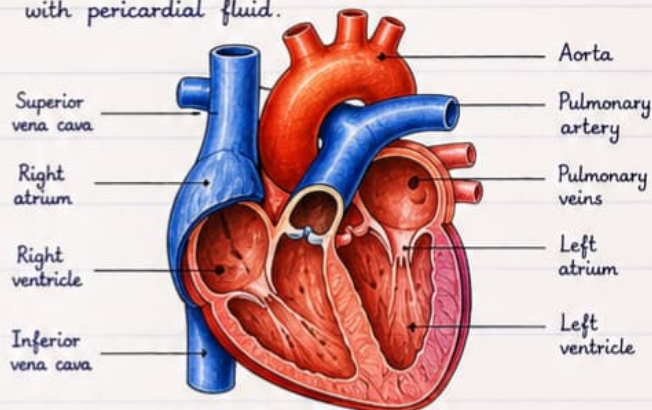


3. Circulatory System

The circulatory system consists of the **heart**, **blood vessels** and **blood**.

Human Heart

- Located in the thoracic cavity between the lungs.
- Muscular, hollow and slightly conical in shape.
- Size ~ fist of the owner.
- Enclosed in a double-walled **pericardium** with pericardial fluid.



Heart is 4-chambered : 2 atria (upper) + 2 ventricles (lower)
 Right side – deoxygenated blood (blue)
 Left side – oxygenated blood (red)

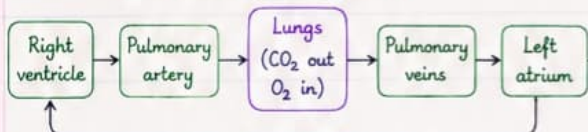
Blood Vessels

Arteries	Veins	Capillaries
• Carry blood away from the heart. • Thick, elastic and muscular walls. • No valves. • High pressure. • Example – Aorta	• Carry blood towards the heart. • Thin walls with less muscle & elasticity. • Have valves. • Low pressure. • Example – Vena cava	• Microscopic vessels connecting arterioles and venules. • Very thin walls (one cell thick). • Site of exchange of materials.
Narrow lumen, thick walls	Wide lumen, thin walls, valves present	One cell thick wall

4. Double Circulation in Humans

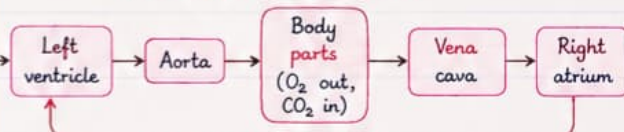
In humans, blood passes through the heart twice during one complete cycle. Hence, it is called **double circulation**.

(A) Pulmonary Circulation



(Carries deoxygenated blood to lungs and brings oxygenated blood to heart)

(B) Systemic Circulation



(Carries oxygenated blood to body and brings deoxygenated blood to heart)

Remember

Oxygenated blood is not always bright red, and deoxygenated blood is not always dark!

Key Summary

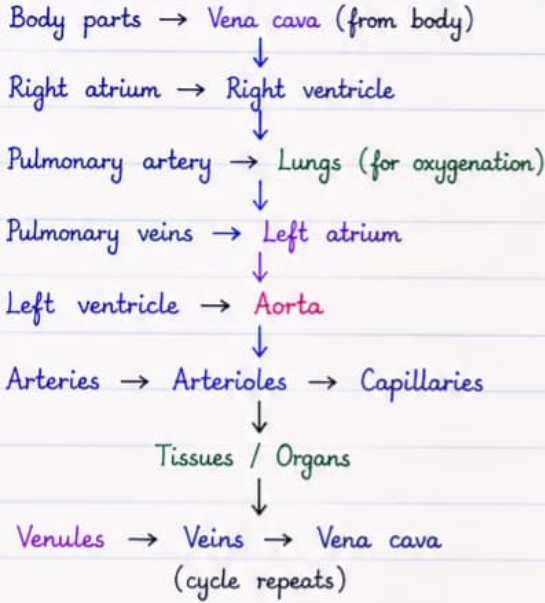
- ✓ Blood is a connective tissue.
- ✓ It has plasma and formed elements.
- ✓ Heart pumps blood through blood vessels.
- ✓ Circulatory system completes transport.
- ✓ Double circulation ensures efficient supply of O_2 to the body.

Did You Know?

Your heart beats about 100,000 times in a day and pumps ~7000 litres of blood!

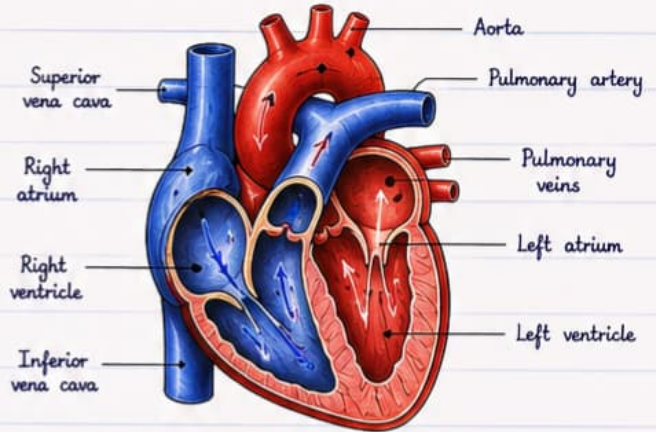
5. Pathway of Blood Flow in Human Heart ♥

Blood flows in a specific pathway through the **heart** and blood vessels to reach all parts of the body.



Important Points ★

- Right side of heart → Deoxygenated blood (blue)
- Left side of heart → Oxygenated blood (red)
- Pulmonary circulation → Heart ↔ Lungs
- Systemic circulation → Heart ↔ Body



→ Deoxygenated blood → Oxygenated blood

6. Cardiac Cycle ♥

One complete heartbeat is called a cardiac cycle. It has three phases:

Phase	Description	Duration	Events
1. Atrial Systole	Atria contract and push blood into ventricles.	~ 0.1 sec	• AV valves open • Semilunar valves closed
2. Ventricular Systole	Ventricles contract and push blood into arteries.	~ 0.3 sec	• AV valves closed • Semilunar valves open
3. General Diastole	All chambers relaxed. Blood flows into atria.	~ 0.4 sec	• All valves closed (momentarily) • Then AV valves open

★ Total time of one cardiac cycle ≈ 0.8 sec
(Heart rate ≈ 72 beats per minute)

8. Heart Sounds 🎵

- "Lub" → First heart sound (closure of AV valves at the beginning of ventricular systole).
- "Dub" → Second heart sound (closure of semilunar valves at the beginning of diastole).
- These sounds can be heard using a stethoscope.

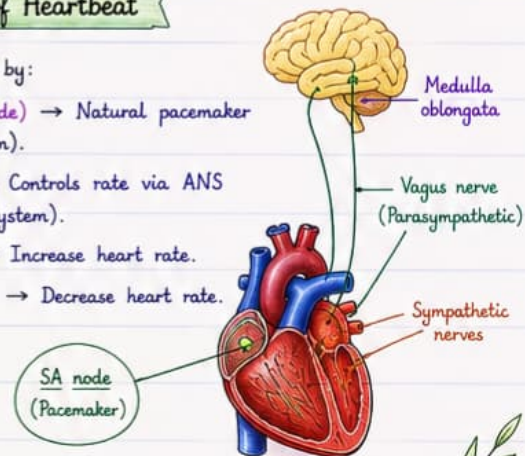
Quick Revision 🌟

1. Blood is a connective tissue.
2. Plasma (55%) + Formed elements (45%).
3. RBCs - transport O₂, WBCs - immunity, Platelets - clotting.
4. Heart is 4-chambered pump.
5. Double circulation maintains efficiency.
6. Cardiac cycle ensures disciplined pumping.
7. ECG helps in detecting heart problems.

9. Regulation of Heartbeat

Heartbeat is regulated by:

- SA node (Sinoatrial node) → Natural pacemaker (located in right atrium).
- Medulla oblongata → Controls rate via ANS (Autonomic Nervous System).
- Sympathetic nerves → Increase heart rate.
- Parasympathetic nerves → Decrease heart rate.



A healthy circulation leads to a healthy life.
Keep your heart strong, keep your life long! ♥♥

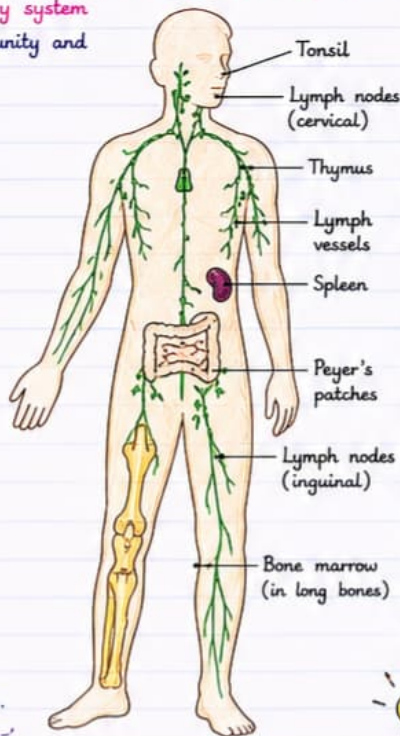
10. Lymphatic System

Lymphatic system is an **accessory circulatory system** that helps in maintaining fluid balance, immunity and returns excess tissue fluid to the blood.

Components

- **Lymph** → A clear, pale yellow fluid similar to plasma but with less proteins and no RBCs.
- **Lymph Capillaries** → Blind ended vessels in tissues that collect excess tissue fluid.
- **Lymph Vessels** → Larger vessels formed by the union of lymph capillaries.
- **Lymph Nodes** → Small, bean shaped structures present at intervals. They filter lymph and trap pathogens.
- **Lymph Organs** → Thymus, Spleen, Tonsils, Peyer's patches, Bone marrow.

Pathway of Lymph



Functions of Lymphatic System

- ★ Drains excess tissue fluid and **returns** it to the bloodstream.
- ★ Absorbs fats and fat soluble vitamins (A, D, E, K) from the small intestine.
- ★ Involved in **immune defence** by producing and transporting lymphocytes.
- ★ Filters lymph through lymph nodes and removes pathogens.

Lymph vs Blood

Lymph	Blood
Colourless / pale yellow	Red
No RBCs, fewer proteins	RBCs present, more proteins
Flows in one direction	Flows in closed circuit
Contains lymphocytes	Contains RBCs, WBCs, platelets



Note: Lymph finally empties into the subclavian veins (near the heart).

11. Disorders of Circulatory System

- **Hypertension** (High Blood Pressure) → Abnormal increase in blood pressure. It puts extra strain on heart.
Causes: Stress, obesity, high salt diet, lack of exercise.
- **Coronary Artery Disease (CAD)** → Narrowing or blockage of coronary arteries due to cholesterol deposits. Leads to heart attack.
- **Angina Pectoris** → Chest pain due to reduced blood flow to heart muscles.
- **Heart Attack** → Sudden blockage of coronary artery leads to death of heart muscles.
- **Stroke** → Interruption of blood supply to brain due to clot or bleeding.
- **Varicose Veins** → Swelling of veins, mostly in legs, due to weak valves.
- **Anaemia** → Deficiency of haemoglobin or RBCs leading to fatigue, weakness.

Prevention is Better than Cure

- ✓ Eat a balanced diet (low fat, high fibre).
- ✓ Exercise regularly (yoga, walking, cycling).
- ✓ Avoid smoking, alcohol and drugs.
- ✓ Maintain normal body weight.
- ✓ Reduce stress and get proper sleep.
- ✓ Regular health check-ups.

"A fit heart is the root of a healthy life."



Quick Summary

Body Fluids

- Water major component.
- Total body fluids ~ 60% of body weight.
- ICF (2/3) inside cells.
- ECF (1/3) outside cells.

Composition of Blood

- Plasma - 55%
- Formed elements - 45%
 - RBCs
 - WBCs
 - Platelets

Circulatory System

- Heart + Blood vessels + Blood Work together to transport materials.
- Heart - 4 chambers, double circulation.

Lymphatic System

- Returns excess tissue fluid to blood.
- Absorbs fats.
- Defends body against pathogens.

Importance

- Maintains homeostasis.
- Distributes nutrients, O₂, hormones.
- Removes wastes.
- Protects from infections.

Remember

- Blood is the river of life.
- Heart is the pump.
- Blood vessels are the roads.
- Lymphatic system is the drainage & defence network.

NEET Tip

Focus on diagrams, flow charts and key differences. Practice NCERT questions and Previous Year Questions.

Did You Know?

An adult heart pumps about 7,000 litres of blood every day and beats around 100,000 times daily!